

C++ Library Management System Project

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In this Library Management System project, you can enter the record of new books and retrieve the details of books available in the library. You can issue the books to the students and maintain their records. Late fine is charged for students who returns the issued books after the due date. Only one book is issued to students. New book is not issued to students those not returned the last book.  [download project](#)



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OPEN

```

//*****
//                                HEADER FILE USED IN PROJECT
//*****

#include<fstream.h>
#include<conio.h>
#include<stdio.h>
#include<process.h>
#include<string.h>
#include<iomanip.h>

//*****
//                                CLASS USED IN PROJECT
//*****

class book
{
    char bno[6];
    char bname[50];
    char aname[20];
public:
    void create_book()
    {
        cout<<"\nNEW BOOK ENTRY...\n";
        cout<<"\nEnter The book no.";
        cin>>bno;
        cout<<"\nEnter The Name of The Book ";
        gets(bname);
        cout<<"\nEnter The Author's Name ";
        gets(aname);
        cout<<"\n\nBook Created..";
    }

    void show_book()
    {
        cout<<"\nBook no. : "<<bno;
        cout<<"\nBook Name : ";
        puts(bname);
        cout<<"Author Name : ";
        puts(aname);
    }

    void modify_book()
    {
        cout<<"\nBook no. : "<<bno;
        cout<<"\nModify Book Name : ";
        gets(bname);
        cout<<"\nModify Author's Name of Book : ";
        gets(aname);
    }

    char* retbno()
    {
        return bno;
    }

    void report()
    {cout<<bno<<setw(30)<<bname<<setw(30)<<aname<<endl;}

};          //class ends here

class student
{

```

```

    char admno[6];
    char name[20];
    char stbno[6];

    int token;
public:
    void create_student()
    {
        clrscr();
        cout<<"\nNEW STUDENT ENTRY...\n";
        cout<<"\nEnter The admission no. ";
        cin>>admno;
        cout<<"\n\nEnter The Name of The Student ";
        gets(name);
        token=0;
        stbno[0]='\0';
        cout<<"\n\nStudent Record Created..";
    }

    void show_student()
    {
        cout<<"\nAdmission no. : "<<admno;
        cout<<"\nStudent Name : ";
        puts(name);
        cout<<"\nNo of Book issued : "<<token;
        if(token==1)
            cout<<"\nBook No "<<stbno;
    }

    void modify_student()
    {
        cout<<"\nAdmission no. : "<<admno;
        cout<<"\nModify Student Name : ";
        gets(name);
    }

    char* retadmno()
    {
        return admno;
    }

    char* retstbno()
    {
        return stbno;
    }

    int rettoken()
    {
        return token;
    }

    void addtoken()
    {token=1;}

    void resettoken()
    {token=0;}

    void getstbno(char t[])
    {
        strcpy(stbno,t);
    }

    void report()
    {cout<<"\t"<<admno<<setw(20)<<name<<setw(10)<<token<<endl;}

};          //class ends here

```

```

//*****
//          global declaration for stream object, object
.....

```

```

//*****

fstream fp,fp1;
book bk;
student st;

//*****
//      function to write in file
//*****

void write_book()
{
    char ch;
    fp.open("book.dat",ios::out|ios::app);
    do
    {
        clrscr();
        bk.create_book();
        fp.write((char*)&bk,sizeof(book));
        cout<<"\n\nDo you want to add more record..(y/n?)";
        cin>>ch;
    }while(ch=='y' || ch=='Y');
    fp.close();
}

void write_student()
{
    char ch;
    fp.open("student.dat",ios::out|ios::app);
    do
    {
        st.create_student();
        fp.write((char*)&st,sizeof(student));
        cout<<"\n\nDo you want to add more record..(y/n?)";
        cin>>ch;
    }while(ch=='y' || ch=='Y');
    fp.close();
}

//*****
//      function to read specific record from file
//*****

void display_spb(char n[])
{
    cout<<"\nBOOK DETAILS\n";
    int flag=0;
    fp.open("book.dat",ios::in);
    while(fp.read((char*)&bk,sizeof(book)))
    {
        if(strcmpi(bk.retbn(),n)==0)
        {
            bk.show_book();
            flag=1;
        }
    }

    fp.close();
    if(flag==0)
        cout<<"\n\nBook does not exist";
    getch();
}

void display_sps(char n[])
{
    cout<<"\nSTUDENT DETAILS\n";
    int flag=0;
    fp.open("student.dat",ios::in);
    while(fp.read((char*)&st,sizeof(student)))
    {

```

```

    {
        if((strcmpi(st.retadmno(),n)==0))
        {
            st.show_student();
            flag=1;
        }
    }

    fp.close();
    if(flag==0)
        cout<<"\n\nStudent does not exist";
    getch();
}

//*****
//      function to modify record of file
//*****

void modify_book()
{
    char n[6];
    int found=0;
    clrscr();
    cout<<"\n\n\tMODIFY BOOK RECORD... ";
    cout<<"\n\n\tEnter The book no. of The book";
    cin>>n;
    fp.open("book.dat",ios::in|ios::out);
    while(fp.read((char*)&bk,sizeof(book)) && found==0)
    {
        if(strcmpi(bk.retbno(),n)==0)
        {
            bk.show_book();
            cout<<"\n\nEnter The New Details of book"<<endl;
            bk.modify_book();
            int pos=-1*sizeof(bk);
            fp.seekp(pos,ios::cur);
            fp.write((char*)&bk,sizeof(book));
            cout<<"\n\n\tRecord Updated";
            found=1;
        }
    }

    fp.close();
    if(found==0)
        cout<<"\n\nRecord Not Found ";
    getch();
}

void modify_student()
{
    char n[6];
    int found=0;
    clrscr();
    cout<<"\n\n\tMODIFY STUDENT RECORD... ";
    cout<<"\n\n\tEnter The admission no. of The student";
    cin>>n;
    fp.open("student.dat",ios::in|ios::out);
    while(fp.read((char*)&st,sizeof(student)) && found==0)
    {
        if(strcmpi(st.retadmno(),n)==0)
        {
            st.show_student();
            cout<<"\n\nEnter The New Details of student"<<endl;
            st.modify_student();
            int pos=-1*sizeof(st);
            fp.seekp(pos,ios::cur);
            fp.write((char*)&st,sizeof(student));
            cout<<"\n\n\tRecord Updated";
            found=1;
        }
    }
}

```

```

    }
}

fp.close();
if(found==0)
    cout<<"\n\n Record Not Found ";
getch();
}

//*****
//      function to delete record of file
//*****

void delete_student()
{
    char n[6];
    int flag=0;
    clrscr();
    cout<<"\n\n\n\tDELETE STUDENT...";
    cout<<"\n\nEnter The admission no. of the Student You Want To Delete : ";
    cin>>n;
    fp.open("student.dat",ios::in|ios::out);
    fstream fp2;
    fp2.open("Temp.dat",ios::out);
    fp.seekg(0,ios::beg);
    while(fp.read((char*)&st,sizeof(student)))
    {
        if(strcmpi(st.retadmno(),n)!=0)
            fp2.write((char*)&st,sizeof(student));
        else
            flag=1;
    }

    fp2.close();
    fp.close();
    remove("student.dat");
    rename("Temp.dat", "student.dat");
    if(flag==1)
        cout<<"\n\n\tRecord Deleted ..";
    else
        cout<<"\n\nRecord not found";
    getch();
}

void delete_book()
{
    char n[6];
    clrscr();
    cout<<"\n\n\n\tDELETE BOOK ...";
    cout<<"\n\nEnter The Book no. of the Book You Want To Delete : ";
    cin>>n;
    fp.open("book.dat",ios::in|ios::out);
    fstream fp2;
    fp2.open("Temp.dat",ios::out);
    fp.seekg(0,ios::beg);
    while(fp.read((char*)&bk,sizeof(book)))
    {
        if(strcmpi(bk.retbno(),n)!=0)
        {
            fp2.write((char*)&bk,sizeof(book));
        }
    }

    fp2.close();
    fp.close();
    remove("book.dat");
    rename("Temp.dat", "book.dat");
    cout<<"\n\n\tRecord Deleted ..";
    getch();
}

```

```

//*****
//      function to display all students list
//*****

void display_allb()
{
    clrscr();
    fp.open("student.dat",ios::in);
    if(!fp)
    {
        cout<<"ERROR!!! FILE COULD NOT BE OPEN ";
        getch();
        return;
    }

    cout<<"\n\n\t\tSTUDENT LIST\n\n";
    cout<<"===== \n";
    cout<<"\tAdmission No."<<setw(10)<<"Name"<<setw(20)<<"Book Issued\n";
    cout<<"===== \n";

    while(fp.read((char*)&st,sizeof(student)))
    {
        st.report();
    }

    fp.close();
    getch();
}

//*****
//      function to display Books list
//*****

void display_allb()
{
    clrscr();
    fp.open("book.dat",ios::in);
    if(!fp)
    {
        cout<<"ERROR!!! FILE COULD NOT BE OPEN ";
        getch();
        return;
    }

    cout<<"\n\n\t\tBook LIST\n\n";

    cout<<"===== \n";

    cout<<"Book Number"<<setw(20)<<"Book Name"<<setw(25)<<"Author\n";

    cout<<"===== \n";

    while(fp.read((char*)&bk,sizeof(book)))
    {
        bk.report();
    }

    fp.close();
    getch();
}

//*****
//      function to issue book
//*****

void book_issue()
.

```

```

{
    char sn[6],bn[6];
    int found=0,flag=0;
    clrscr();
    cout<<"\n\nBOOK ISSUE ...";
    cout<<"\n\n\tEnter The student's admission no.";
    cin>>sn;
    fp.open("student.dat",ios::in|ios::out);
    fp1.open("book.dat",ios::in|ios::out);
    while(fp.read((char*)&st,sizeof(student)) && found==0)
    {
        if(strcmpi(st.retadmno(),sn)==0)
        {
            found=1;
            if(st.rettoken()==0)
            {
                cout<<"\n\n\tEnter the book no. ";
                cin>>bn;
                while(fp1.read((char*)&bk,sizeof(book))&& flag==0)
                {
                    if(strcmpi(bk.retbno(),bn)==0)
                    {
                        bk.show_book();
                        flag=1;
                        st.addtoken();
                        st.getstbno(bk.retbno());
                        int pos=-1*sizeof(st);
                        fp.seekp(pos,ios::cur);
                        fp.write((char*)&st,sizeof(student));
                        cout<<"\n\n\t Book issued successfully\n\nPlease Note:
Write current date
in backside of book and submit within 15 days fine Rs. 1
for each day
after 15 days period";
                    }
                }
                if(flag==0)
                    cout<<"Book no does not exist";
            }
            else
                cout<<"You have not returned the last book ";
        }
    }
    if(found==0)
        cout<<"Student record not exist...";
    getch();
    fp.close();
    fp1.close();
}

```

```

//*****
//      function to deposit book
//*****

```

```

void book_deposit()
{
    char sn[6],bn[6];
    int found=0,flag=0,day,fine;
    clrscr();
    cout<<"\n\nBOOK DEPOSIT ...";
    cout<<"\n\n\tEnter The student's admission no.";
    cin>>sn;
    fp.open("student.dat",ios::in|ios::out);
    fp1.open("book.dat",ios::in|ios::out);
    while(fp.read((char*)&st,sizeof(student)) && found==0)
    {
        if(strcmpi(st.retadmno(),sn)==0)
        {
            found=1;
            if(st.rettoken()==1)
            {

```



```

while(tp1.read((char*)&bk,sizeof(book))&& flag==0)
{
    if(strcmpi(bk.retbno(),st.retstbno())==0)
    {
        bk.show_book();
        flag=1;
        cout<<"\n\nBook deposited in no. of days";
        cin>>day;
        if(day>15)
        {
            fine=(day-15)*1;
            cout<<"\n\nFine has to deposited Rs. "<<fine;
        }
        st.resettoken();
        int pos=-1*sizeof(st);
        fp.seekp(pos,ios::cur);
        fp.write((char*)&st,sizeof(student));
        cout<<"\n\n\t Book deposited successfully";
    }
}
if(flag==0)
    cout<<"Book no does not exist";
}
else
    cout<<"No book is issued..please check!!";
}
}
if(found==0)
    cout<<"Student record not exist...";
getch();
fp.close();
fp1.close();
}

```

```

//*****
//      INTRODUCTION FUNCTION
//*****

```

```

void intro()
{
    clrscr();
    gotoxy(35,11);
    cout<<"LIBRARY";
    gotoxy(35,14);
    cout<<"MANAGEMENT";
    gotoxy(35,17);
    cout<<"SYSTEM";
    cout<<"\n\nMADE BY : YOUR NAME";
    cout<<"\n\nSCHOOL : SCHOOL NAME";
    getch();
}

```

```

//*****
//      ADMINISTRATOR MENU FUNCTION
//*****

```

```

void admin_menu()
{
    clrscr();
    int ch2;
    cout<<"\n\n\n\tADMINISTRATOR MENU";
    cout<<"\n\n\t1.CREATE STUDENT RECORD";
    cout<<"\n\n\t2.DISPLAY ALL STUDENTS RECORD";
    cout<<"\n\n\t3.DISPLAY SPECIFIC STUDENT RECORD ";
    cout<<"\n\n\t4.MODIFY STUDENT RECORD";
    cout<<"\n\n\t5.DELETE STUDENT RECORD";
    cout<<"\n\n\t6.CREATE BOOK ";
    //*****
}

```

```

cout<<"\n\n\t7.DISPLAY ALL BOOKS ";
cout<<"\n\n\t8.DISPLAY SPECIFIC BOOK ";
cout<<"\n\n\t9.MODIFY BOOK ";
cout<<"\n\n\t10.DELETE BOOK ";
cout<<"\n\n\t11.BACK TO MAIN MENU";
cout<<"\n\n\tPlease Enter Your Choice (1-11) ";
cin>>ch2;
switch(ch2)
{
    case 1: clrscr();
        write_student();break;
    case 2: display_all();break;
    case 3:
        char num[6];
        clrscr();
        cout<<"\n\n\tPlease Enter The Admission No. ";
        cin>>num;
        display_sps(num);
        break;
    case 4: modify_student();break;
    case 5: delete_student();break;
    case 6: clrscr();
        write_book();break;
    case 7: display_allb();break;
    case 8: {
        char num[6];
        clrscr();
        cout<<"\n\n\tPlease Enter The book No. ";
        cin>>num;
        display_spb(num);
        break;
    }
    case 9: modify_book();break;
    case 10: delete_book();break;
    case 11: return;
    default:cout<<"\a";
}
admin_menu();
}

//*****
//      THE MAIN FUNCTION OF PROGRAM
//*****

void main()
{
    char ch;
    intro();
    do
    {
        clrscr();
        cout<<"\n\n\n\tMAIN MENU";
        cout<<"\n\n\t01. BOOK ISSUE";
        cout<<"\n\n\t02. BOOK DEPOSIT";
        cout<<"\n\n\t03. ADMINISTRATOR MENU";
        cout<<"\n\n\t04. EXIT";
        cout<<"\n\n\tPlease Select Your Option (1-4) ";
        ch=getche();
        switch(ch)
        {
            case '1':clrscr();
                book_issue();
                break;
            case '2':book_deposit();
                break;
            case '3':admin_menu();
                break;
            case '4':exit(0);
            default :cout<<"\a";
        }
    }
}

```

```
}while(ch!='4');  
}  
  
//*****  
//      END OF PROJECT  
//*****
```

 

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